

INDIAN INSTITUTE OF TECHNOLOGY PATNA



EC3202

DIGITAL SIGNAL PROCESSING

EXPERIMENT 05: IIR Filter Design - Analog Filter

SUBMITTED BY:

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Aim 1: Design a low-pass Butterworth filter to satisfy

Passband cutoff: $\Omega_p = 0.2\pi$; Passband ripple: $R_p = 7$ dB

Stopband cutoff: $\Omega_s = 0.3\pi$; Stopband ripple: $A_s = 16$ dB

MATLAB Code -

```
clc; clear; close all;
```

```
wp = 0.2;
```

```
ws = 0.3;
```

```
Rp = 7;
```

```
As = 16;
```

```
[N, Wn] = buttord(wp, ws, Rp, As);
```

```
[b, a] = butter(N, Wn, 'low');
```

```
disp(['Filter Order N = ', num2str(N)]);
```

```
disp(['Cutoff Frequency Wn = ', num2str(Wn)]);
```

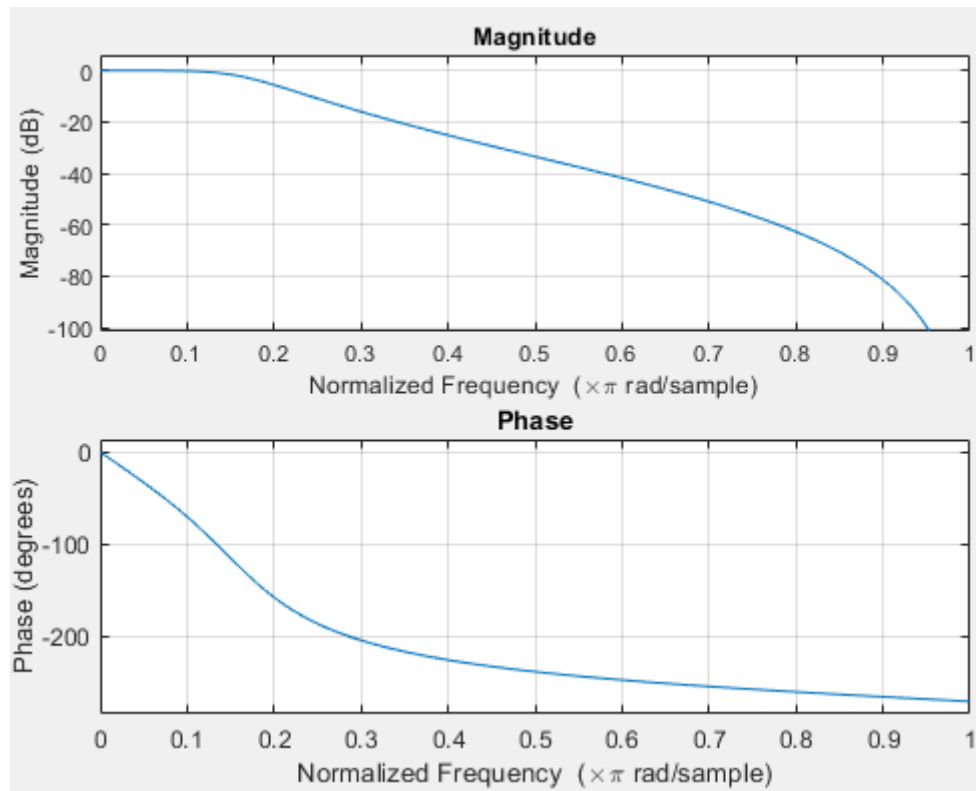
```
freqz(b, a, 1024);
```

Output and Plot :

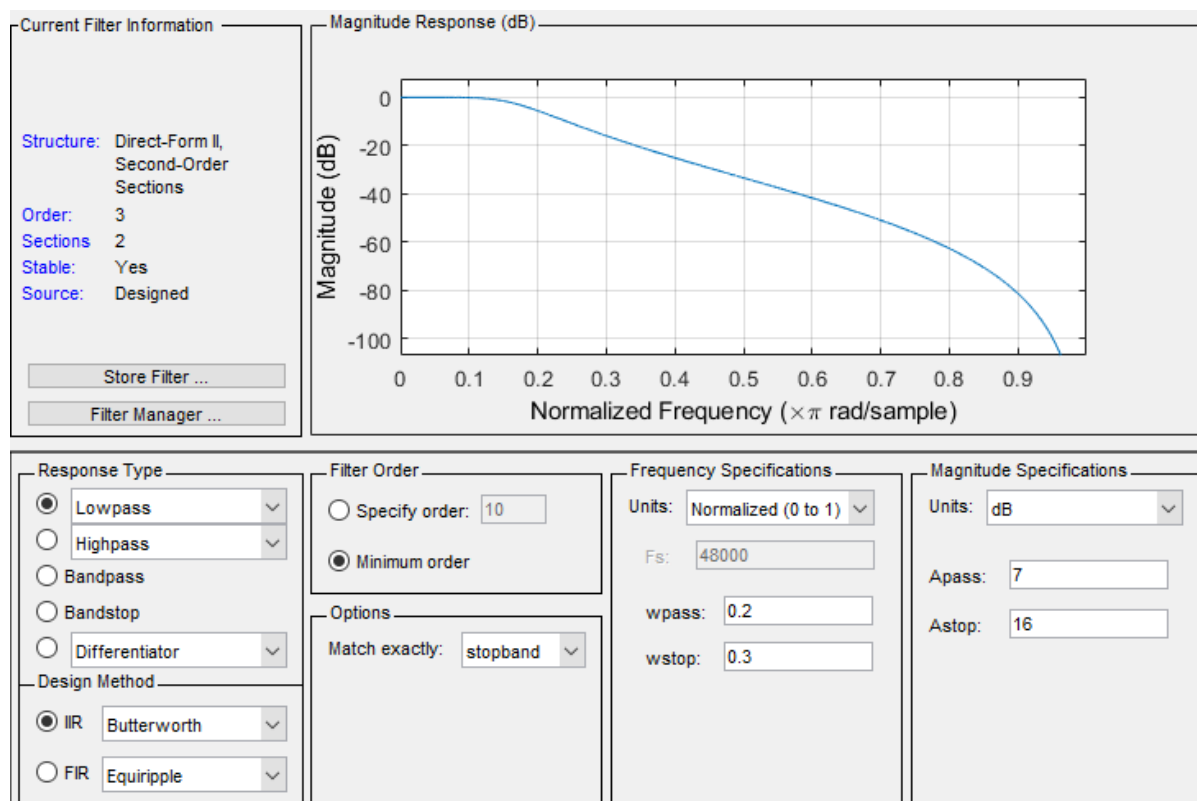
Command Window

```
Filter Order N = 3
```

```
Cutoff Frequency Wn = 0.17198
```



Using Filter Designer :



Aim 2: Design a lowpass Chebyshev-I filter to satisfy

Passband cutoff: $\Omega_p = 0.2\pi$; Passband ripple: $R_p = 1$ dB

Stopband cutoff: $\Omega_s = 0.3\pi$; Stopband ripple: $A_s = 16$ dB

MATLAB Code :

```
clc; clear; close all;
```

```
wp = 0.2;
```

```
ws = 0.3;
```

```
Rp = 1;
```

```
As = 16;
```

```
[N, Wn] = cheb1ord(wp, ws, Rp, As);
```

```
[b, a] = cheby1(N, Rp, Wn, 'low');
```

```
disp(['Filter Order N = ', num2str(N)]);
```

```
disp(['Cutoff Frequency Wn = ', num2str(Wn)]);
```

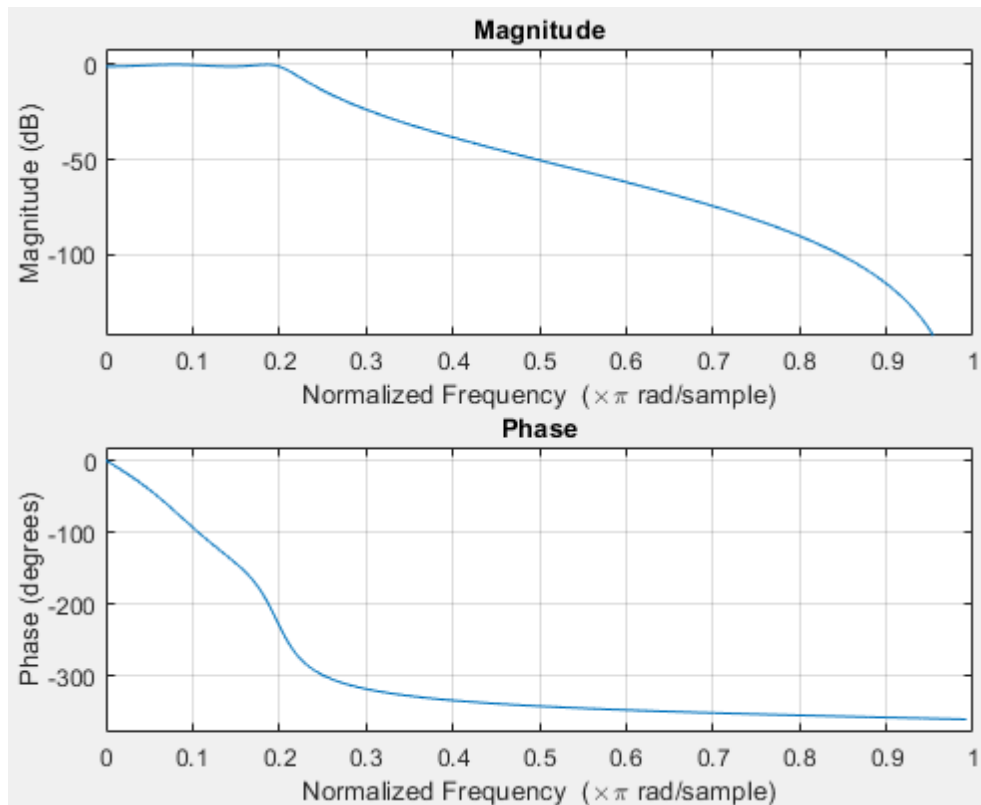
```
freqz(b, a, 1024);
```

Output and Plot :

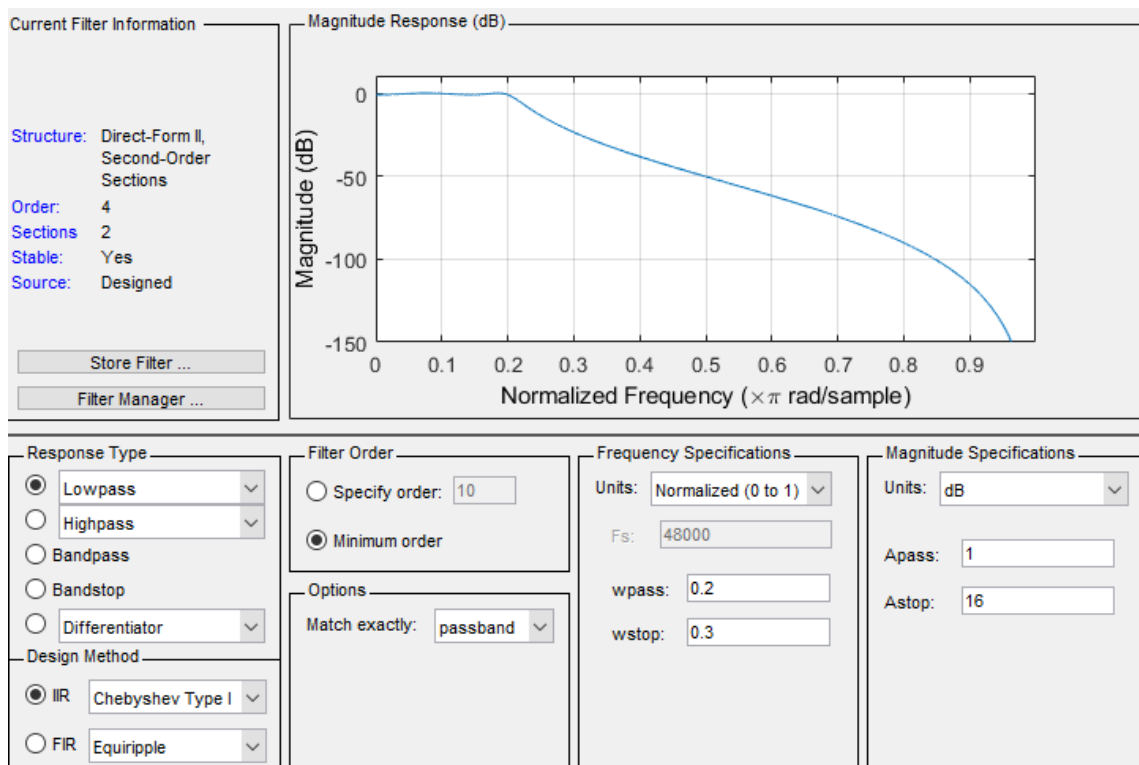
Command Window

```
Filter Order N = 4
```

```
Cutoff Frequency Wn = 0.2
```



Using Filter Designer :



Aim 3: Design a low-pass digital filter (Butterworth) using impulse invariance technique to satisfy –

$$\omega_p = 0.2\pi, \quad R_p = 1 \text{ dB}$$

$$\omega_s = 0.3\pi, \quad A_s = 15 \text{ dB}$$

MATLAB Code -

```
clc; clear; close all;
```

```
wp = 0.2 * pi;
```

```
ws = 0.3 * pi;
```

```
Rp = 1;
```

```
As = 15;
```

```
T = 1;
```

```
Omega_p = wp / T;
```

```
Omega_s = ws / T;
```

```
[N, Omega_c] = buttord(Omega_p, Omega_s, Rp, As, 's');
```

```
[b, a] = butter(N, Omega_c, 's');
```

```
[bd, ad] =impinvar(b, a, 1/T);
```

```
disp(['Filter Order N = ', num2str(N)]);
```

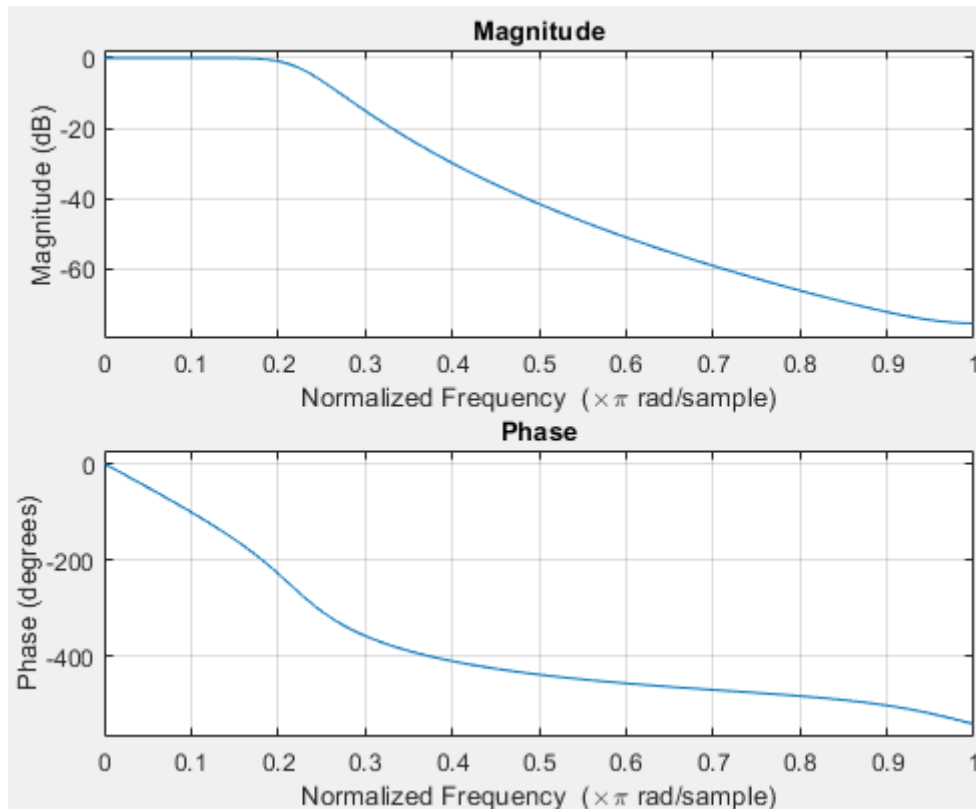
```
disp(['Cutoff Frequency Omega_c = ', num2str(Omega_c)]);
```

```
freqz(bd, ad, 1024);
```

Output and Plot :

Command Window

```
Filter Order N = 6  
Cutoff Frequency Omega_c = 0.70865
```



Aim 4: Design a lowpass digital filter (Chebyshev-I) using impulse invariance technique to satisfy - $\omega_p = 0.2\pi$, $R_p = 1$ dB

$$\omega_s = 0.3\pi, \quad A_s = 15 \text{ dB}$$

MATLAB Code -

```
clc; clear; close all;
```

```
wp = 0.2 * pi;
```

```
ws = 0.3 * pi;
```

```
Rp = 1;
```

```
As = 15;
```

```
T = 1;
```

```
Omega_p = wp / T;
```

```
Omega_s = ws / T;
```

```
[N, Omega_p_new] = cheb1ord(Omega_p, Omega_s, Rp, As, 's');
```

```
[b, a] = cheby1(N, Rp, Omega_p_new, 's');
```

```
[bd, ad] =impinvar(b, a, 1/T);
```

```
disp(['Filter Order N = ', num2str(N)]);
```

```
disp(['Adjusted Passband Frequency = ', num2str(Omega_p_new)]);
```

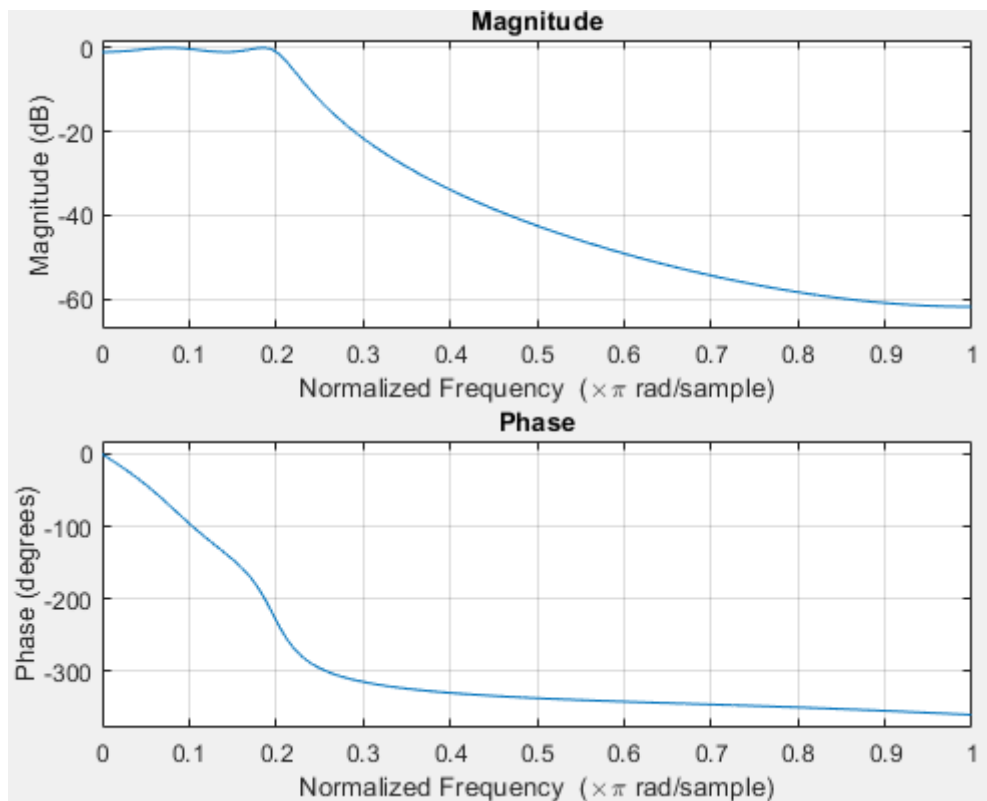
```
freqz(bd, ad, 1024);
```

Output and Plot :

Command Window

```
Filter Order N = 4
```

```
Adjusted Passband Frequency = 0.62832
```



Aim 5: Design a digital filter (Butterworth) using Bilinear transformation technique to satisfy - $\omega_p = 0.2\pi$, $R_p = 1$ dB

$$\omega_s = 0.3\pi, \quad A_s = 15 \text{ dB}$$

MATLAB Code -

```
clc; clear; close all;
```

```
wp = 0.2 * pi;
```

```
ws = 0.3 * pi;
```

```
Rp = 1;
```

```
As = 15;
```

```
T = 1;
```

```
Omega_p = (2/T) * tan(wp/2);
```

```
Omega_s = (2/T) * tan(ws/2);
```

```
[N, Omega_c] = buttord(Omega_p, Omega_s, Rp, As, 's');
```

```
[b, a] = butter(N, Omega_c, 's');
```

```
[bd, ad] = bilinear(b, a, 1/T);
```

```
disp(['Filter Order N = ', num2str(N)]);
```

```
disp(['Cutoff Frequency Omega_c = ', num2str(Omega_c)]);
```

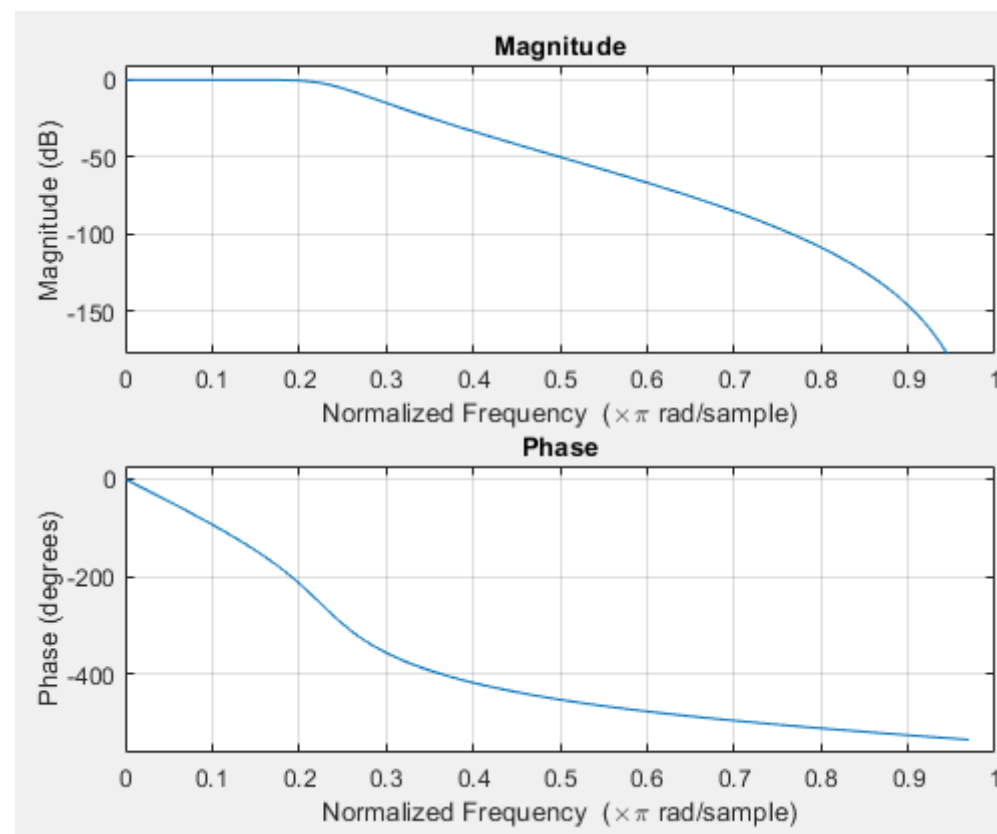
```
freqz(bd, ad, 1024);
```

Output and Plot :

Command Window

```
Filter Order N = 6
```

```
Cutoff Frequency Omega_c = 0.76623
```



Aim 6: Design a digital filter (Chebyshev-I) using Bilinear transformation technique to satisfy - $\omega_p = 0.2\pi$, $R_p = 1$ dB

$$\omega_s = 0.3\pi, \quad A_s = 15 \text{ dB}$$

MATLAB Code :

```
clc; clear; close all;
```

```
wp = 0.2 * pi;
```

```
ws = 0.3 * pi;
```

```
Rp = 1;
```

```
As = 15;
```

```
T = 1;
```

```
Omega_p = (2/T) * tan(wp/2);
```

```
Omega_s = (2/T) * tan(ws/2);
```

```
[N, Omega_p_new] = cheb1ord(Omega_p, Omega_s, Rp, As, 's');
```

```
[b, a] = cheby1(N, Rp, Omega_p_new, 's');
```

```
[bd, ad] = bilinear(b, a, 1/T);
```

```
disp(['Filter Order N = ', num2str(N)]);
```

```
disp(['Adjusted Passband Frequency = ', num2str(Omega_p_new)]);
```

```
freqz(bd, ad, 1024);
```

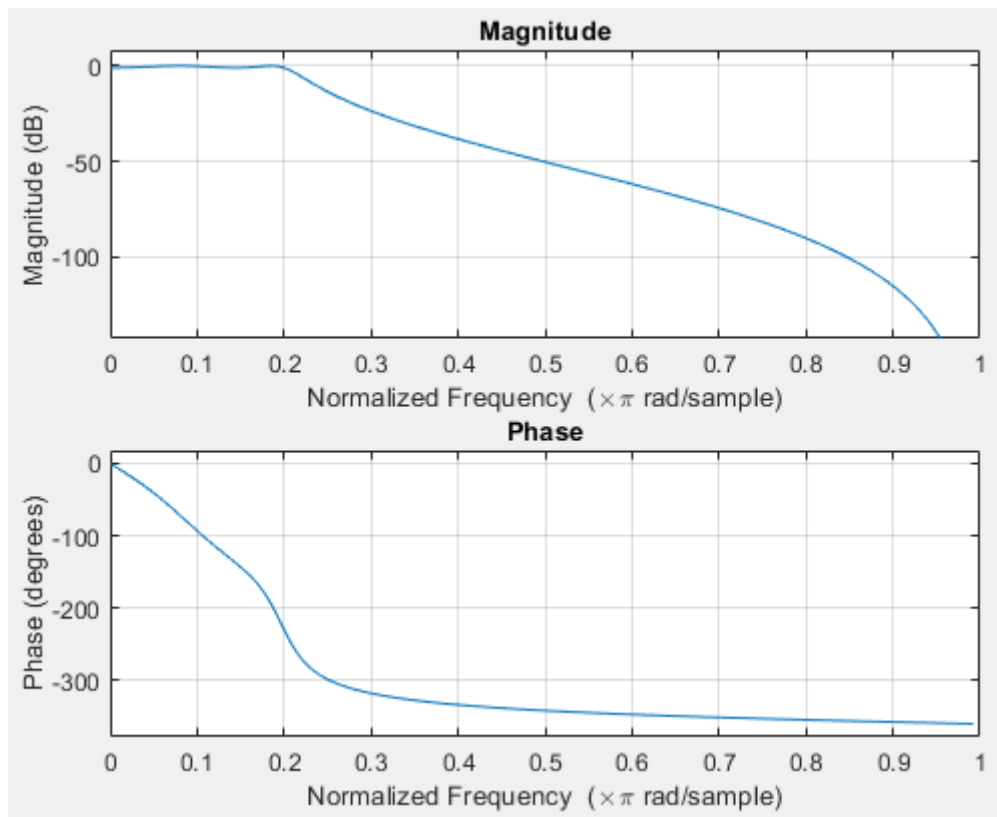
Output and Plot :

Command Window

Filter Order N = 4

Adjusted Passband Frequency = 0.64984

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Conclusion :

In this experiment, low-pass digital filters were designed using both Butterworth and Chebyshev Type-I approaches. Different design methods were applied, such as direct digital design, the impulse invariance technique, and the bilinear transformation method.

The observations show that Butterworth filters produce a smooth and flat passband response, while Chebyshev filters achieve a steeper transition band but introduce ripple in the passband. The impulse invariance method was seen to maintain the time-domain behavior of the system, although it results in aliasing in the frequency domain. On the other hand, the bilinear transformation approach avoids aliasing and ensures a one-to-one correspondence between analog and digital frequencies.